```
!pip install spacy
```

```
/lib/python3.12/dist-packages (3.8.11)
>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
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sr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
l/lib/python3.12/dist-packages (from spacy) (3.1.6)
local/lib/python3.12/dist-packages (from spacy) (75.2.0)
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6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy)
.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (
4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spac
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4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.4)
r/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)
in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)
in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2026.2.25)
in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.3)
0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.1.5)
sr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (0.24.1)
>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (0.23.0)
5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.5.1)
/usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.4.2->spacy) (2.1.1)
r/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (8.3.1)
in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (
r/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy) (13.9.4
2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0->spacy)
.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim<1.0.0,
13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim<1.0.
local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer
```

```
!python -m spacy download en_core_web_sm
```

```
spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 69.7 MB/s eta 0:00:00

('en_core_web_sm')

you may need to restart Python in
s. You can do this by selecting the
n.
```

```
import spacy

# Load English model
nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully")
```

```
spaCy model loaded successfully
```

```
sentences = [
"The student reads a book.",
```

```
"The dog chased the cat.",
"She writes code every day.",
"The teacher explained the lesson clearly.",
"John bought a new laptop.",
"The sun rises in the east.",
"The engineer designed a new system.",
"The meeting was scheduled by the manager.",
"They completed the project successfully.",
"The scientist discovered a new element.",
"The team won the match yesterday.",
"The computer processes data quickly.",
"The report was written by the analyst.",
"Can you solve this problem?",
"Although it was raining, they continued the journey."
]

print("Total sentences:", len(sentences))
```

```
Total sentences: 15
```

```
for sentence in sentences:
    doc = nlp(sentence)

    print("\nSentence:", sentence)
    print("Token\tLemma\tPOS\tDependency\tHead")

    for token in doc:
        print(token.text, "\t", token.lemma_, "\t", token.pos_, "\t", token.dep_, "\t", token.hea
```

```
Sentence: The student reads a book.
Token   Lemma   POS     Dependency      Head
The      the     DET     det     student
student          student         NOUN    nsubj   reads
reads    read    VERB    ROOT    reads
a        a       DET     det     book
book     book    NOUN    dobj    reads
.        .       PUNCT   punct   reads

Sentence: The dog chased the cat.
Token   Lemma   POS     Dependency      Head
The      the     DET     det     dog
dog      dog     NOUN    nsubj   chased
chased   chase   VERB    ROOT    chased
the      the     DET     det     cat
cat      cat     NOUN    dobj    chased
.        .       PUNCT   punct   chased

Sentence: She writes code every day.
Token   Lemma   POS     Dependency      Head
She      she     PRON    nsubj   writes
writes   write   VERB    ROOT    writes
code     code    NOUN    dobj    writes
every    every   DET     det     day
day      day     NOUN    npadvmod        writes
.        .       PUNCT   punct   writes

Sentence: The teacher explained the lesson clearly.
Token   Lemma   POS     Dependency      Head
The      the     DET     det     teacher
teacher          teacher         NOUN    nsubj   explained
explained        explain         VERB    ROOT    explained
the      the     DET     det     lesson
lesson   lesson          NOUN    dobj    explained
clearly          clearly         ADV     advmod          explained
.        .       PUNCT   punct   explained

Sentence: John bought a new laptop.
Token   Lemma   POS     Dependency      Head
John     John    PROPN   nsubj   bought
```

```
bought   buy     VERB    ROOT    bought
a        a       DET     det     laptop
new      new     ADJ     amod    laptop
laptop   laptop          NOUN    dobj    bought
.        .       PUNCT   punct   bought

Sentence: The sun rises in the east.
Token   Lemma   POS     Dependency      Head
The      the     DET     det     sun
sun      sun     NOUN    nsubj   rises
rises    rise    VERB    ROOT    rises
in       in      ADP     prep    rises
the      the     DET     det     east
east     east    NOUN    pobj    in
.        .       PUNCT   punct   rises
```

```
import pandas as pd

sentence = "The student reads a book."
doc = nlp(sentence)

data = []

for token in doc:
    data.append({
        "Token": token.text,
        "Lemma": token.lemma_,
        "POS": token.pos_,
        "Dependency": token.dep_,
        "Head": token.head.text
    })

df = pd.DataFrame(data)
df
```

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	student
1	student	student	NOUN	nsubj	reads
2	reads	read	VERB	ROOT	reads
3	a	a	DET	det	book
4	book	book	NOUN	dobj	reads
5	.	.	PUNCT	punct	reads

Next steps: Generate code with df New interactive sheet

```
sentence = "The student reads a book."
doc = nlp(sentence)

subject = ""
verb = ""
obj = ""

for token in doc:
    if token.dep_ == "nsubj":
        subject = token.text
    if token.dep_ == "ROOT":
        verb = token.text
    if token.dep_ == "dobj":
        obj = token.text

print("Subject:", subject)
```

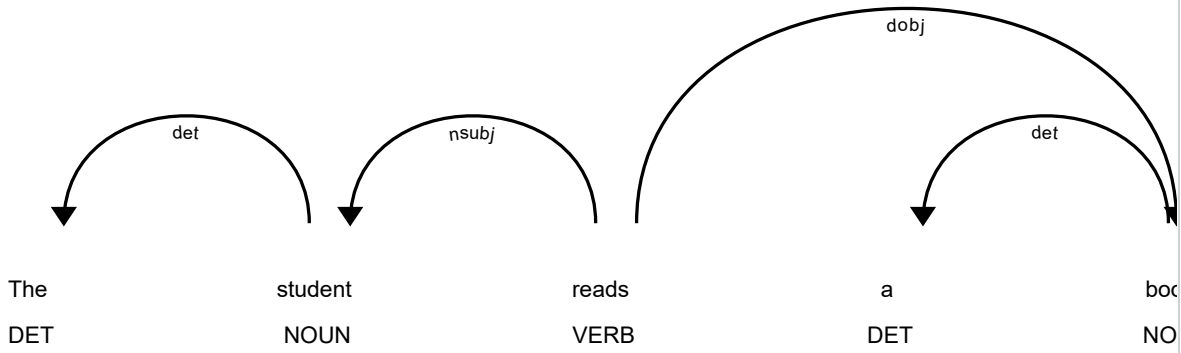
```
print("Verb:", verb)
print("Object:", obj)
```

```
Subject: student
Verb: reads
Object: book
```

```
from spacy import displacy
```

```
sentence = "The student reads a book."
doc = nlp(sentence)

displacy.render(doc, style="dep", jupyter=True)
```



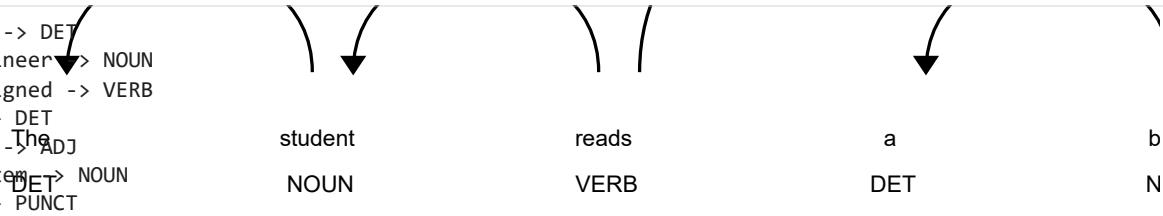

```
sentences_to_visualize = [
"The student reads a book.",
"The dog chased the cat.",
"The meeting was scheduled by the manager.",
"Can you solve this problem?",
"Although it was raining, they continued the journey."
]

for sentence in sentences_to_visualize:
    doc = nlp(sentence)
    displacy.render(doc, style="dep", jupyter=True)
```

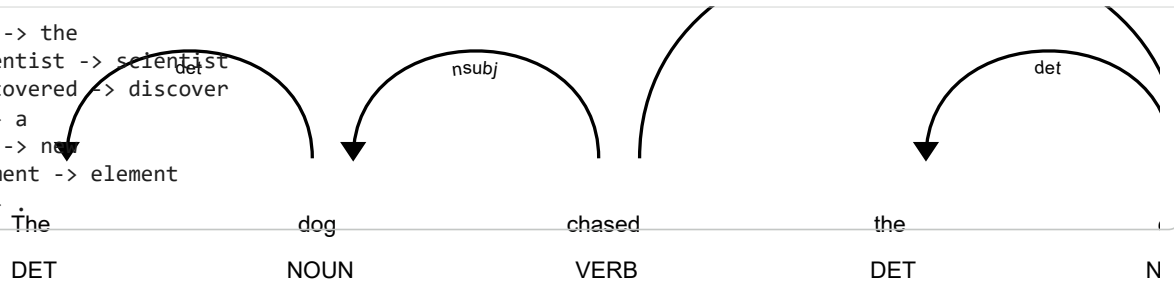
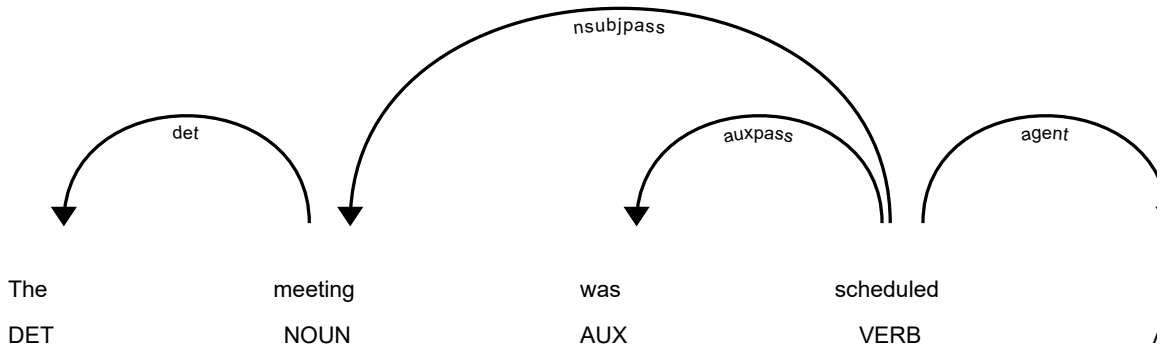
```
sentence = "The engineer designed a new system."
doc = nlp(sentence)

for token in doc:
    print(token.text, "->", token.pos_)
```

```
The -> DET
engineer -> NOUN
designed -> VERB
a -> DET
new -> ADJ
system -> NOUN
. -> PUNCT
```




```
sentence = "The scientist discovered a new element."
doc = nlp(sentence)

for token in doc:
    print(token.text, "->", token.lemma_)
```

```
The -> the
scientist -> scientist
discovered -> discover
a -> a
new -> new
element -> element
. -> .
```